

AI Engineer

Step 4: Build AI Projects

CareerCompass — Detailed Learning Notes

What this step is about

Apply ML/DL knowledge to complete, reproducible AI projects.

Core Concepts

- Choose a measurable problem such as classification, forecasting, recommendation, text classification or image recognition.
- Define success before choosing a model. Start with a baseline.
- Track datasets, preprocessing, experiments, model versions and evaluation.
- Write tests for important data transformations and inference code.

Skills You Should Be Able To Demonstrate

- Define an objective metric.
- Create a baseline.
- Track experiments.
- Build reproducible training/inference code.
- Document limitations and failure cases.

Hands-on Project / Practice

- Build an end-to-end AI project using a public dataset.
- Include data preparation, baseline, trained model, evaluation report and a simple demo/API.

Recommended References

- Kaggle — <https://www.kaggle.com/>
- Machine Learning in 2024 — <https://www.youtube.com/watch?v=bmmQA8A-yUA>
- Machine Learning with Python and Scikit-Learn — <https://www.youtube.com/watch?v=hDKCxebp88A>